CSE400_L17_Scribe

Generate a lecture scribe intended strictly as exam-oriented reference material for the course CSE 400: Fundamentals of Probability in Computing. The scribe must be a faithful and exam-ready reconstruction of the lecture and must be written so that a student can rely on it as the sole reference while preparing for a closed-notes or reading-based examination.
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❑ Why Study Joint Random Variables?
❑ Discrete Case
• Joint PMF Table Representation
• Example: Rolling Two Dice
❑ Continuous Case
• Geometric Interpretation
• Worked Example
❑ Joint Cumulative Distribution Function
• Definition of Joint CDF
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• Discrete Case
• Properties of the Joint CDF
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CSE 400: Fundamentals of Probability in Computing

Lecture Scribe — Joint Random Variables

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1. Why Study Joint Random Variables

A random variable describes the outcome of a random experiment numerically. In many situations, a single random variable is insufficient to describe the behavior of a system. Multiple quantities may be random simultaneously, and their behavior must be studied together.

Joint random variables allow the simultaneous modeling of two or more random quantities arising from the same experiment. They enable the study of how random variables behave together rather than individually.

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2. Discrete Case

2.1 Joint PMF Table Representation

For two discrete random variables X and Y, the **joint probability mass function (joint PMF)** specifies the probability that X and Y simultaneously take particular values.

The joint PMF is defined as

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

The joint PMF may be represented in tabular form, where

- Rows correspond to values of X
- Columns correspond to values of Y
- Each entry represents the joint probability $P(X = x, Y = y)$

The probabilities satisfy the condition

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

2.2 Example: Rolling Two Dice

Consider the experiment of rolling two fair six-sided dice.

Define two random variables:

$$X = \text{Outcome of the first die}$$

$$Y = \text{Outcome of the second die}$$

Each die can take values

$$1, 2, 3, 4, 5, 6$$

Since the dice are fair and independent, each ordered pair (x, y) has probability

$$P(X = x, Y = y) = \frac{1}{36}$$

because there are $6 \times 6 = 36$ equally likely outcomes.

Joint PMF Table

X \ Y	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36

X \ Y	1	2	3	4	5	6
6	1/36	1/36	1/36	1/36	1/36	1/36

Verification of Total Probability

$$\sum_{x=1}^{6} \sum_{y=1}^{6} P(X = x, Y = y)$$

$$= 36 \times \frac{1}{36}$$

$$= 1$$

Thus the joint PMF satisfies the normalization condition.

---

3. Continuous Case

For continuous random variables, probabilities are described using a **joint probability density function (joint PDF)**.

$$f_{X,Y}(x, y)$$

represents the density of the random variables X and Y.

The probability that (X, Y) lies within a region R is

$$P((X, Y) \in R) = \int \int_R f_{X,Y}(x, y)\, dx\, dy$$

The density function satisfies

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$$

3.1 Geometric Interpretation

In the continuous case, the joint PDF can be visualized geometrically.

- The joint PDF defines a surface above the xy-plane.
- The probability of a region corresponds to the **volume under the surface** over that region.

Thus,

$$P((X, Y) \in R)$$

is equal to the volume under the density surface above region R.

3.2 Worked Example

Consider two continuous random variables X and Y with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} c & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Step 1: Determine c

Since the total probability must equal 1,

$$\int_0^1 \int_0^1 c\, dx\, dy = 1$$

First integrate with respect to x:

$$\int_0^1 c\, dx = c$$

Then integrate with respect to y:

$$\int_0^1 c\, dy = c$$

Thus

$$c = 1$$

Therefore

$$f_{X,Y}(x, y) = \begin{cases} 1 & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

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4. Joint Cumulative Distribution Function

4.1 Definition of Joint CDF

For two random variables X and Y, the **joint cumulative distribution function (joint CDF)** is defined as

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

The joint CDF gives the probability that both random variables are less than or equal to specified values.

4.2 Continuous Case

If X and Y are continuous random variables with joint PDF $f_{X,Y}(x, y)$, then

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, dv\, du$$

This expression accumulates the density over the region

$$u \leq x, \quad v \leq y$$

4.3 Discrete Case

If X and Y are discrete random variables with joint PMF $p_{X,Y}(x, y)$, then the joint CDF is

$$F_{X,Y}(x, y) = \sum_{u \leq x} \sum_{v \leq y} p_{X,Y}(u, v)$$

The sums include all pairs (u, v) such that

$$u \leq x \quad \text{and} \quad v \leq y$$

4.4 Properties of the Joint CDF

The joint cumulative distribution function satisfies the following properties.

Property 1: Range

$$0 \leq F_{X,Y}(x, y) \leq 1$$

Property 2: Nondecreasing Property

$F_{X,Y}(x, y)$ is nondecreasing in both arguments.

If

$$x_1 \leq x_2$$

then

$$F_{X,Y}(x_1, y) \leq F_{X,Y}(x_2, y)$$

Similarly, if

$$y_1 \leq y_2$$

then

$$F_{X,Y}(x, y_1) \leq F_{X,Y}(x, y_2)$$

Property 3: Limits

$$\lim_{x \to -\infty} F_{X,Y}(x, y) = 0$$

$$\lim_{y \to -\infty} F_{X,Y}(x, y) = 0$$

$$\lim_{x \to \infty, y \to \infty} F_{X,Y}(x, y) = 1$$

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5. Comprehensive Example

Consider the discrete random variables X and Y obtained from rolling two fair dice.

Each outcome (x, y) has probability

$$P(X = x, Y = y) = \frac{1}{36}$$

Compute the Joint CDF

The joint CDF is

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

Suppose

$$x = 2, \quad y = 3$$

Then

$$F_{X,Y}(2,3) = P(X \le 2,\ Y \le 3)$$

Possible values:

$$X = 1, 2$$

$$Y = 1, 2, 3$$

Number of valid ordered pairs:

$$2 \times 3 = 6$$

Each outcome has probability

$$\frac{1}{36}$$

Thus

$$F_{X,Y}(2,3) = 6 \times \frac{1}{36}$$

$$= \frac{6}{36}$$

$$= \frac{1}{6}$$

Final Result of Example

$$F_{X,Y}(2,3) = \frac{1}{6}$$

This value represents the probability that the first die is at most 2 and the second die is at most 3.

End of Lecture Scribe

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Lecture Scribe — Joint Random Variables

---

1. Why Study Joint Random Variables

A random variable assigns a numerical value to the outcome of a random experiment. In many experiments, more than one quantity may be random simultaneously. In such cases, a single random variable is not sufficient to represent the outcomes of the experiment. Instead, multiple random variables are defined and studied together.

Joint random variables allow the study of the behavior of two or more random variables simultaneously. This framework enables the analysis of probabilities associated with multiple random quantities that arise from the same experiment.

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2. Discrete Case

2.1 Joint PMF Table Representation

Let X and Y be two discrete random variables. The **joint probability mass function (joint PMF)** of X and Y is defined as

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

This function specifies the probability that the random variable X takes the value x and the random variable Y takes the value y simultaneously.

The joint PMF can be represented using a table. In such a table:

- Each row corresponds to a possible value of X.
- Each column corresponds to a possible value of Y.
- Each entry of the table contains the probability $P(X = x, Y = y)$.

The probabilities in the joint PMF must satisfy the normalization condition

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

where the sums are taken over all possible values of X and Y.

2.2 Example: Rolling Two Dice

Consider an experiment in which two fair six-sided dice are rolled.

Define the following random variables:

$$X = \text{outcome of the first die}$$

$$Y = \text{outcome of the second die}$$

Each die can take one of the values

$$1, 2, 3, 4, 5, 6$$

Since both dice are fair and all outcomes are equally likely, the total number of ordered outcomes is

$$6 \times 6 = 36$$

Each ordered pair (x, y) therefore has probability

$$P(X = x, Y = y) = \frac{1}{36}$$

Joint PMF Table

$X \backslash Y$	**1**	**2**	**3**	**4**	**5**	**6**
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36

$X \backslash Y$	**1**	**2**	**3**	**4**	**5**	**6**
6	1/36	1/36	1/36	1/36	1/36	1/36

Verification of Normalization Condition

The sum of all joint probabilities must equal 1.

$$\sum_{x=1}^{6} \sum_{y=1}^{6} P(X = x, Y = y)$$

Since there are 36 entries in the table and each equals $1/36$,

$$36 \times \frac{1}{36} = 1$$

Thus the normalization condition is satisfied.

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3. Continuous Case

For continuous random variables X and Y, the probability distribution is described using the **joint probability density function (joint PDF)**.

The joint PDF is denoted as

$$f_{X,Y}(x, y)$$

The probability that the pair (X, Y) lies within a region R is given by

$$P((X, Y) \in R) = \int \int_R f_{X,Y}(x, y)\, dx\, dy$$

The joint PDF must satisfy the condition

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$$

3.1 Geometric Interpretation

In the continuous case, the joint PDF can be interpreted geometrically.

The function $f_{X,Y}(x, y)$ forms a surface above the xy-plane. The probability that the pair (X, Y) lies in a region R corresponds to the **volume under this surface above the region** R.

Thus,

$$P((X, Y) \in R)$$

is equal to the volume under the surface $f_{X,Y}(x, y)$ above region R.

3.2 Worked Example

Consider the joint probability density function

$$f_{X,Y}(x, y) = \begin{cases} c & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Step 1: Determine the constant c

The total probability must equal 1. Therefore,

$$\int_0^1 \int_0^1 c\, dx\, dy = 1$$

First integrate with respect to x:

$$\int_0^1 c\, dx = c(1 - 0) = c$$

Then integrate with respect to y:

$$\int_0^1 c\, dy = c(1 - 0) = c$$

Thus,

$$c = 1$$

Therefore the joint PDF becomes

$$f_{X,Y}(x, y) = \begin{cases} 1 & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

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4. Joint Cumulative Distribution Function

4.1 Definition of Joint CDF

For random variables X and Y , the **joint cumulative distribution function (joint CDF)** is defined as

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

The joint CDF gives the probability that X is less than or equal to x and Y is less than or equal to y.

4.2 Continuous Case

If X and Y are continuous random variables with joint PDF $f_{X,Y}(x, y)$, then the joint CDF is

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, dv\, du$$

This expression accumulates the probability density over the region

$$u \leq x, \quad v \leq y$$

4.3 Discrete Case

If X and Y are discrete random variables with joint PMF $p_{X,Y}(x, y)$, the joint CDF is given by

$$F_{X,Y}(x, y) = \sum_{u \leq x} \sum_{v \leq y} p_{X,Y}(u, v)$$

The sums include all values of u and v satisfying

$$u \leq x, \quad v \leq y$$

4.4 Properties of the Joint CDF

The joint cumulative distribution function has the following properties.

Property 1: Range

$$0 \le F_{X,Y}(x, y) \le 1$$

Property 2: Nondecreasing Property

The joint CDF is nondecreasing in each argument.

If

$$x_1 \le x_2$$

then

$$F_{X,Y}(x_1, y) \le F_{X,Y}(x_2, y)$$

Similarly, if

$$y_1 \le y_2$$

then

$$F_{X,Y}(x, y_1) \le F_{X,Y}(x, y_2)$$

Property 3: Limits

$$\lim_{x \to -\infty} F_{X,Y}(x, y) = 0$$

$$\lim_{y \to -\infty} F_{X,Y}(x, y) = 0$$

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---

5. Comprehensive Example

Consider the experiment of rolling two fair six-sided dice.

Define

$$X = \text{outcome of the first die}$$

$$Y = \text{outcome of the second die}$$

Each ordered pair (x, y) has probability

$$P(X = x, Y = y) = \frac{1}{36}$$

The joint CDF is defined as

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

Compute $F_{X,Y}(2, 3)$

$$F_{X,Y}(2, 3) = P(X \leq 2, Y \leq 3)$$

Possible values satisfying the condition:

$$X = 1, 2$$

$$Y = 1, 2, 3$$

Total ordered pairs:

$$2 \times 3 = 6$$

Each pair has probability

$$\frac{1}{36}$$

Thus,

$$F_{X,Y}(2, 3) = 6 \times \frac{1}{36} = \frac{6}{36} = \frac{1}{6}$$

Final Result

$$F_{X,Y}(2, 3) = \frac{1}{6}$$

This represents the probability that the first die outcome is less than or equal to 2 and the second die outcome is less than or equal to 3.

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Lecture Scribe — Joint Random Variables

Why Study Joint Random Variables

A random variable assigns a numerical value to the outcome of a random experiment. In many situations, more than one quantity arising from the same experiment is random. In such cases, a single random variable does not fully describe the outcome of the experiment.

To represent multiple random quantities simultaneously, two or more random variables are defined and studied together. The study of these variables together is carried out using **joint random variables**. Joint random variables allow probabilities to be assigned to combinations of outcomes of multiple random variables.

Discrete Case

Joint PMF Table Representation

Let X and Y be two discrete random variables. The **joint probability mass function (joint PMF)** of X and Y is defined as

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

The joint PMF gives the probability that the random variable X takes the value x and the random variable Y takes the value y simultaneously.

The joint PMF can be represented in a table form. In such a table:

- Rows correspond to possible values of X.
- Columns correspond to possible values of Y.
- Each entry represents the probability $P(X = x, Y = y)$.

The probabilities must satisfy the normalization condition

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

where the summations are taken over all possible values of X and Y.

Example: Rolling Two Dice

Consider an experiment in which two fair six-sided dice are rolled.

Define the random variables

$$X = \text{Outcome of the first die}$$

$$Y = \text{Outcome of the second die}$$

Each die can take the values

$$1, 2, 3, 4, 5, 6$$

The total number of ordered outcomes is

$$6 \times 6 = 36$$

Because the dice are fair, each ordered pair (x, y) occurs with probability

$$P(X = x, Y = y) = \frac{1}{36}$$

Joint PMF Table

$X \backslash Y$	**1**	**2**	**3**	**4**	**5**	**6**
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36

$X \backslash Y$	**1**	**2**	**3**	**4**	**5**	**6**
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

Verification of Total Probability

The sum of all entries must equal 1.

$$\sum_{x=1}^{6} \sum_{y=1}^{6} P(X = x, Y = y)$$

There are 36 entries and each equals $1/36$.

$$36 \times \frac{1}{36} = 1$$

Thus the normalization condition is satisfied.

Continuous Case

For continuous random variables X and Y, the joint distribution is described using a **joint probability density function (joint PDF)**.

The joint PDF is denoted as

$$f_{X,Y}(x, y)$$

The probability that the pair (X, Y) lies within a region R is given by

$$P((X, Y) \in R) = \int \int_R f_{X,Y}(x, y)\, dx\, dy$$

The joint PDF must satisfy

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$$

Geometric Interpretation

In the continuous case, the joint PDF can be interpreted geometrically.

The function $f_{X,Y}(x, y)$ forms a surface above the xy-plane. The probability that the random pair (X, Y) lies in a region R is equal to the **volume under the surface** $f_{X,Y}(x, y)$ above that region.

Thus,

$$P((X, Y) \in R)$$

corresponds to the volume under the surface above region R.

Worked Example

Consider the joint probability density function

$$f_{X,Y}(x, y) = \begin{cases} c & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Step 1: Determine c

The total probability must equal 1.

$$\int_0^1 \int_0^1 c\, dx\, dy = 1$$

First integrate with respect to x:

$$\int_0^1 c\, dx = c(1 - 0) = c$$

Then integrate with respect to y:

$$\int_0^1 c\, dy = c(1 - 0) = c$$

Thus,

$$c = 1$$

Therefore,

$$f_{X,Y}(x,y) = \begin{cases} 1 & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Joint Cumulative Distribution Function

Definition of Joint CDF

For random variables X and Y, the **joint cumulative distribution function (joint CDF)** is defined as

$$F_{X,Y}(x,y) = P(X \le x, Y \le y)$$

The joint CDF gives the probability that the random variable X is less than or equal to x and the random variable Y is less than or equal to y.

Continuous Case

If X and Y are continuous random variables with joint PDF $f_{X,Y}(x,y)$, the joint CDF is

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u,v)\, dv\, du$$

This expression accumulates the probability density over the region

$$u \le x, \quad v \le y$$

Discrete Case

If X and Y are discrete random variables with joint PMF $p_{X,Y}(x,y)$, the joint CDF is

$$F_{X,Y}(x,y) = \sum_{u \le x} \sum_{v \le y} p_{X,Y}(u,v)$$

The sums include all values satisfying

$$u \le x, \quad v \le y$$

Properties of the Joint CDF

Property 1: Range

$$0 \leq F_{X,Y}(x, y) \leq 1$$

Property 2: Nondecreasing Property

The joint CDF is nondecreasing in each argument.

If

$$x_1 \leq x_2$$

then

$$F_{X,Y}(x_1, y) \leq F_{X,Y}(x_2, y)$$

If

$$y_1 \leq y_2$$

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Property 3: Limits

$$\lim_{x \to -\infty} F_{X,Y}(x, y) = 0$$

$$\lim_{y \to -\infty} F_{X,Y}(x, y) = 0$$

$$\lim_{x \to \infty, y \to \infty} F_{X,Y}(x, y) = 1$$

Comprehensive Example

Consider the experiment of rolling two fair six-sided dice.

Define

$$X = \text{Outcome of the first die}$$

$$Y = \text{Outcome of the second die}$$

Each ordered pair (x, y) occurs with probability

$$P(X = x, Y = y) = \frac{1}{36}$$

The joint cumulative distribution function is

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

Compute $F_{X,Y}(2, 3)$

$$F_{X,Y}(2, 3) = P(X \leq 2, Y \leq 3)$$

Possible values satisfying the conditions:

$$X = 1, 2$$

$$Y = 1, 2, 3$$

The number of ordered pairs is

$$2 \times 3 = 6$$

Each pair has probability

$$\frac{1}{36}$$

Thus,

$$F_{X,Y}(2, 3) = 6 \times \frac{1}{36} = \frac{6}{36} = \frac{1}{6}$$

Result

$$F_{X,Y}(2, 3) = \frac{1}{6}$$

This value represents the probability that the outcome of the first die is less than or equal to 2 and the outcome of the second die is less than or equal to 3.